



ST BENEDICT'S

SUBJECT	Physics IebT
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Hilder
DATE	15 July 2025
MARKS	20
DURATION	30 minutes

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 10 multiple-choice questions on 7 pages (including this front page). A Data Sheet of 2 pages (i-ii) is provided. Please check that your question paper is complete.
2. Fill in the circles completely with HB pencil
3. Please colour on the appropriate circle: ☐ A ☐ B ☒ C ☐ D
Please DO NOT: ☐ ☒ ☒ ☒
Put a line or cross or partially colour in.
To make a change, erase completely and fill in the correct circle.
4. Read all the questions carefully.
5. Answer all the questions.
6. **No calculators may be used in this paper.**
7. All questions are equally weighted with no negative marking.
8. Learners answering this examination on paper will require a B pencil and an eraser.

	Total
Mark	
Moderated Mark	
Question Total	20

Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1

A girl is bouncing on a trampoline

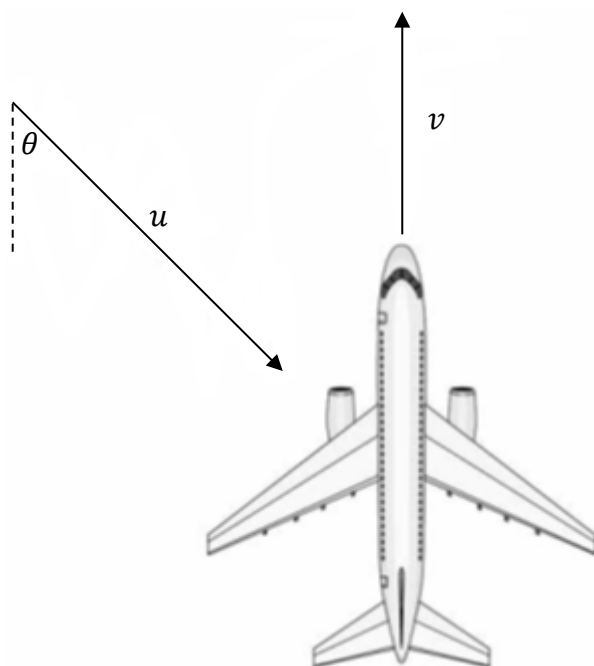
Assuming that air resistance is negligible, her acceleration...

- A is zero when she is at maximum height.
- B is constant when she is in the air.
- C changes direction as she rises and then falls.
- D is maximum just before she lands on the trampoline.

QUESTION 2

An aircraft is flying due north through still air with a speed v

The aircraft enters a region where the wind is blowing with a speed u from a direction which makes an angle of θ with due south.

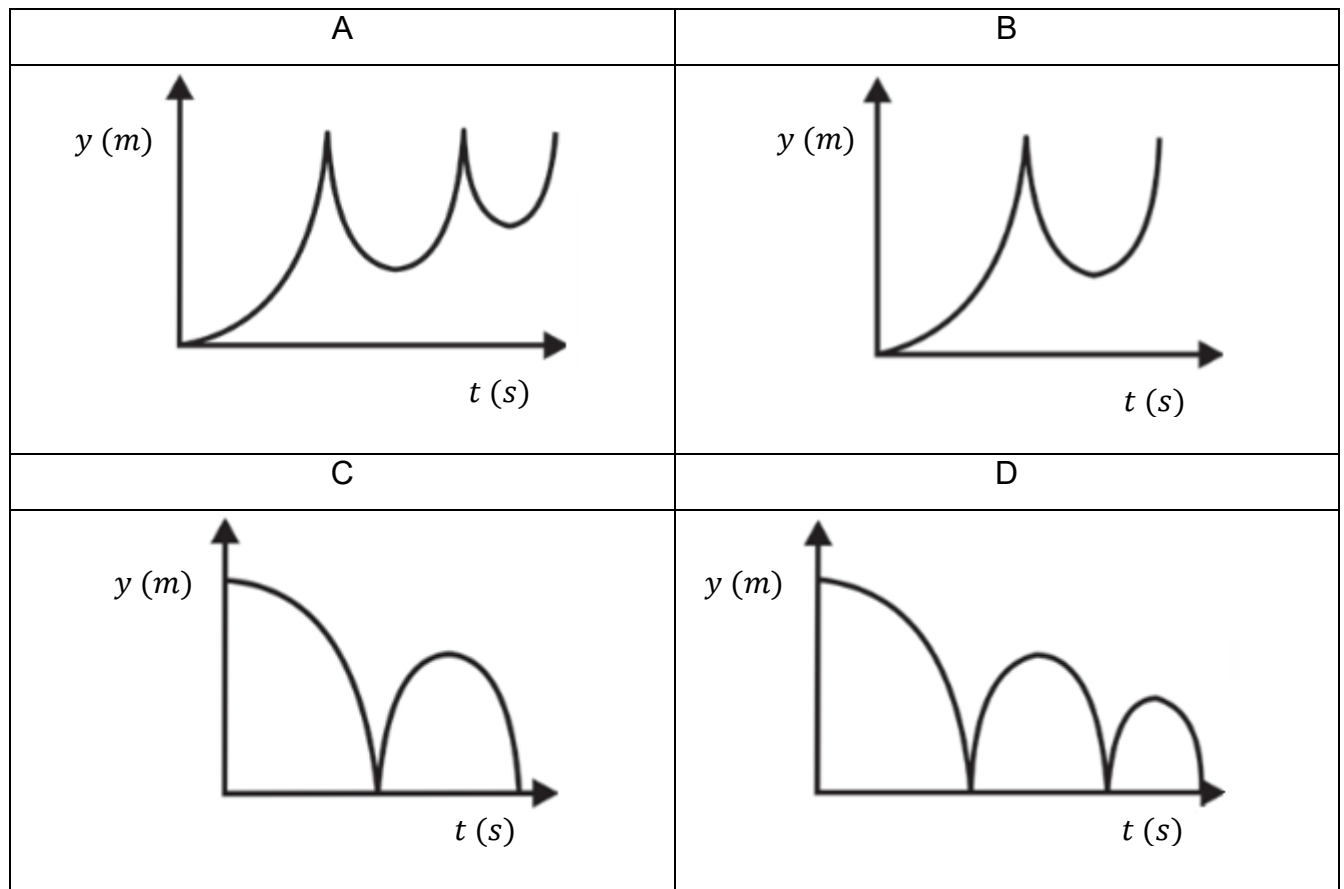


What is the time taken for the aircraft to fly a distance D due north of its current position in this windy region?

- A $\frac{D}{v-u \cos \theta}$
- B $\frac{D}{v-u \sin \theta}$
- C $\frac{D}{v+u \cos \theta}$
- D $\frac{D}{v+u \sin \theta}$

QUESTION 3

Which one of the following position vs time graphs represents a ball that is dropped and bounces TWICE before coming to rest? The frame of reference is the starting point.

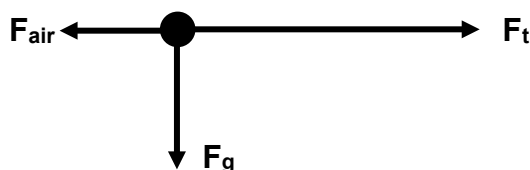
**QUESTION 4**

A ball thrown vertically upwards with an initial velocity v_i reaches a maximum height y . The velocity of the ball when it is halfway up is:

- A $\frac{v_i}{2}$
- B $\sqrt{v_i - 2gy}$
- C $\frac{1}{\sqrt{2}} v_i$
- D gy

QUESTION 5

A student draws the force diagram below showing the forces acting on a ball after it has been thrown. The ball is in mid-flight and is travelling horizontally to the right.



Which force, if any, is **incorrect**?

- A F_{air} , the force due to air resistance
- B F_t , the force due to the throw
- C F_g , the force due to gravity
- D All of the forces are correct

QUESTION 6

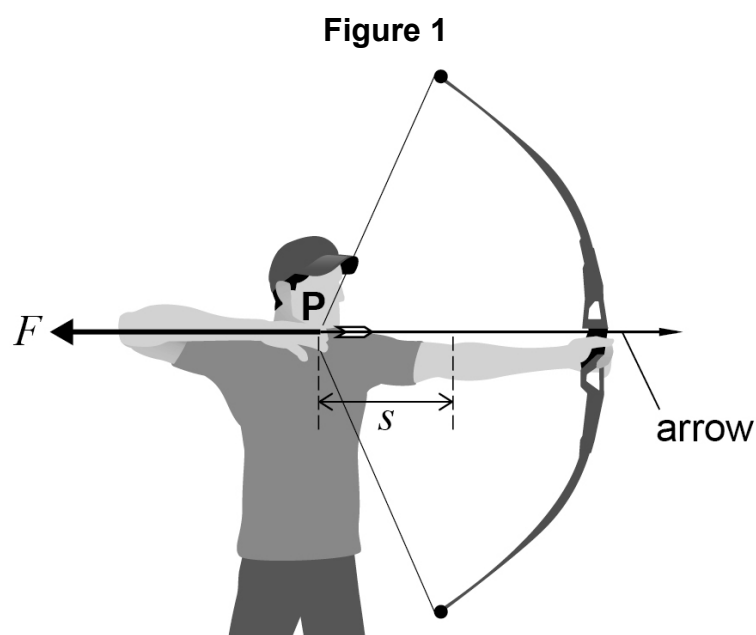
A person weighs 750 N. This person stands on a scale placed on the floor of a lift. The lift travels up and down in a tall building.

Which row in the table below describes the motion of the lift at the moment when the scale reads 700 N?

	Direction of movement	Type of motion
A	Upwards	Constant speed
B	Upwards	Slowing down
C	Downwards	Slowing down
D	Downwards	Constant speed

QUESTION 7

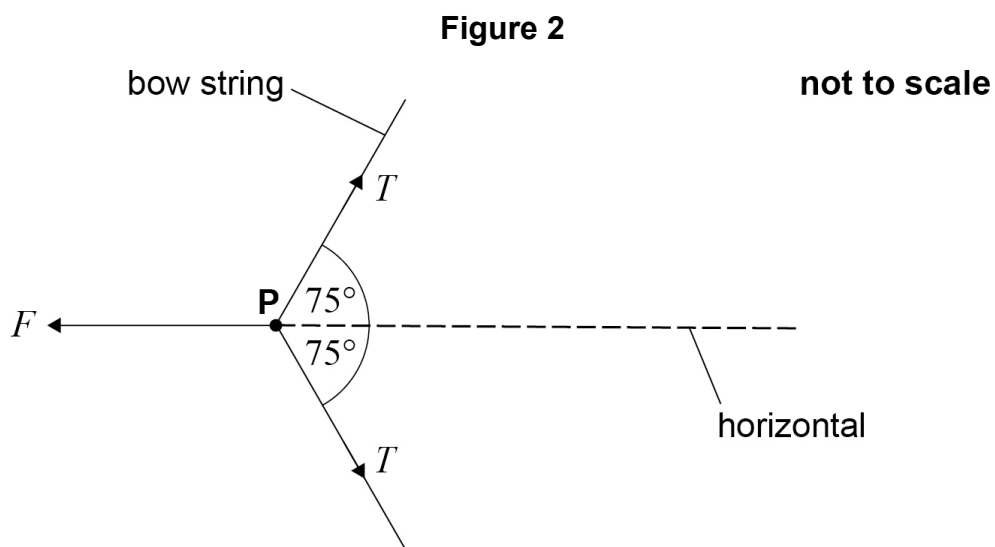
Figure 1 shows an archer using a bow in a competition.



The archer exerts a force F to pull point P on the string back through a distance s .

Figure 2 is a simplified diagram of the bow string showing the forces acting on P.

The tension in the string is T and the string makes an angle of 75° to the horizontal.



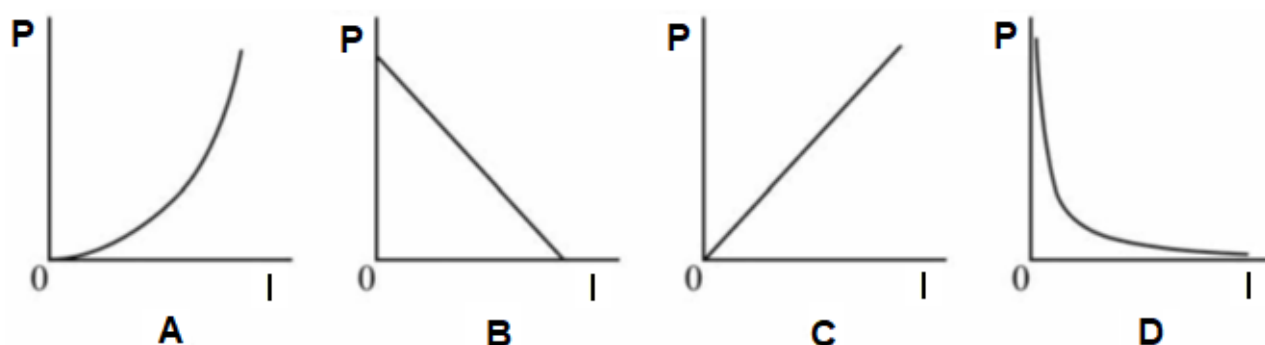
In **Figure 2**, F is 160 N and P is in equilibrium.

Use the information above to determine the correct term of T , the tension in the bow string.

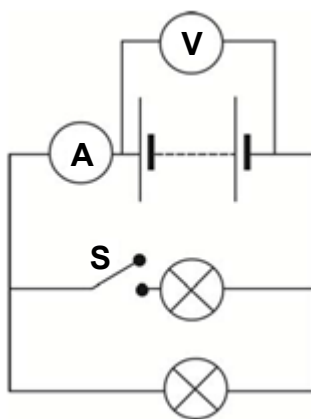
- A $F = T \sin 75^\circ$
- B $F = 2T \sin 75^\circ$
- C $F = T \cos 75^\circ$
- D $F = 2T \cos 75^\circ$

QUESTION 8

Which graph shows how power dissipated P varies with current I in a component that obeys Ohm's law?

**QUESTION 9**

In the circuit shown below the resistance of the ammeter, switch and connecting wires is negligible. The voltmeter has a very high resistance. The battery has **significant** internal resistance.

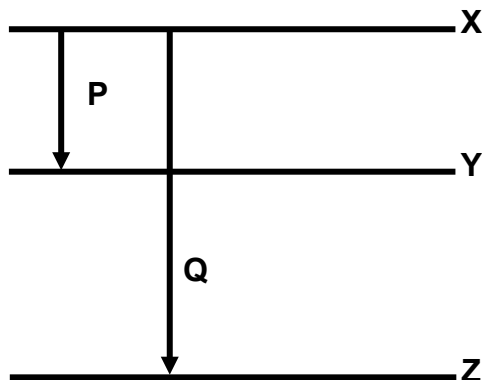


How are the readings on the ammeter and voltmeter affected if switch S is closed?

	AMMETER	VOLTMETER
A	Decrease	Increase
B	Decrease	No change
C	Increase	No change
D	Increase	Decrease

QUESTION 10

The diagram represents 3 energy levels, X, Y and Z, in a certain atom. The energy difference between levels Y and Z is twice the energy difference between levels X and Y. If the wavelength of a photon emitted as a result of transition P, from level X to Y, is λ , then what is the wavelength of the photon emitted during transition Q, from level X to Z?



- A 2λ
- B 3λ
- C $\frac{\lambda}{2}$
- D $\frac{\lambda}{3}$

END OF ASSESSMENT